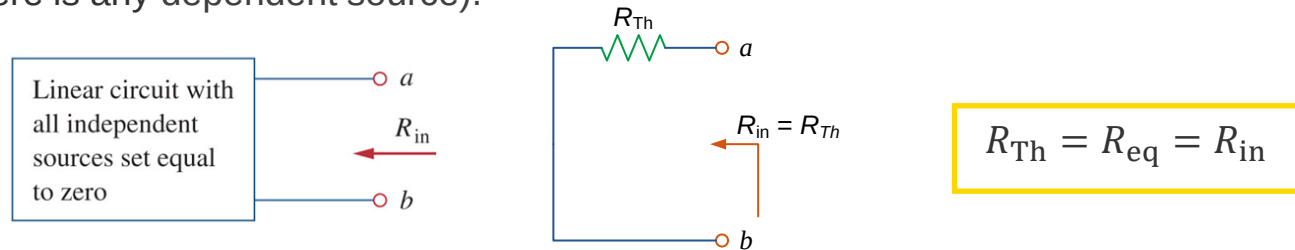


Topic 3 recap

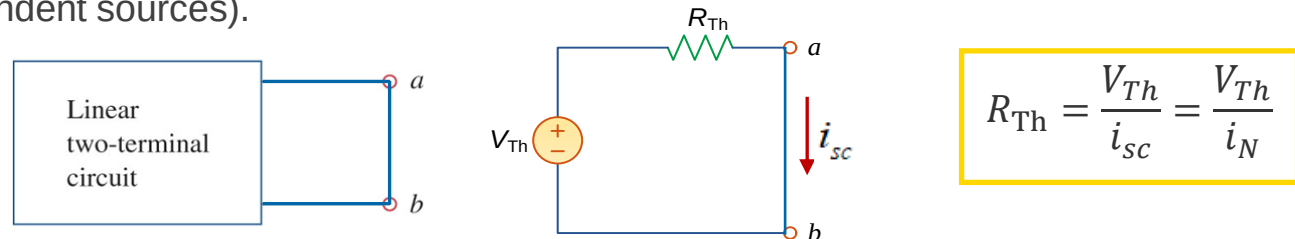
- Superposition principle in linear circuits (circuit analysis).
 - The response of a linear circuit (a voltage or current of an element) with multiple sources is the **algebraic sum** of individual responses **due to each independent source** when the rest are **turned off**.
 - **Dependent sources** are **left intact** during all calculations.
- Source transformation (circuit analysis/simplification).
 - Consists in replacing a **voltage source** v_s (independent or dependent) in **series** with a **resistor** with a **current source** i_s (independent or dependent, respectively) in **parallel** with another resistor at the **same terminal** or vice versa.
 - $i_s = \frac{v_s}{R}$ or $v_s = Ri_s$.
 - The transformed resistors are the **same**.
- Thevenin's theorem.
 - A linear circuit can be **replaced** (modelled) with a **voltage source** V_{Th} in **series** with a **resistor** R_{Th} from a given terminal.
 - V_{Th} is equal to the **open-circuit voltage** across the terminals ($V_{Th} = v_{oc}$).
 - 3 options to obtain R_{Th} depending on the source types (independent/dependent).

Topic 3 recap

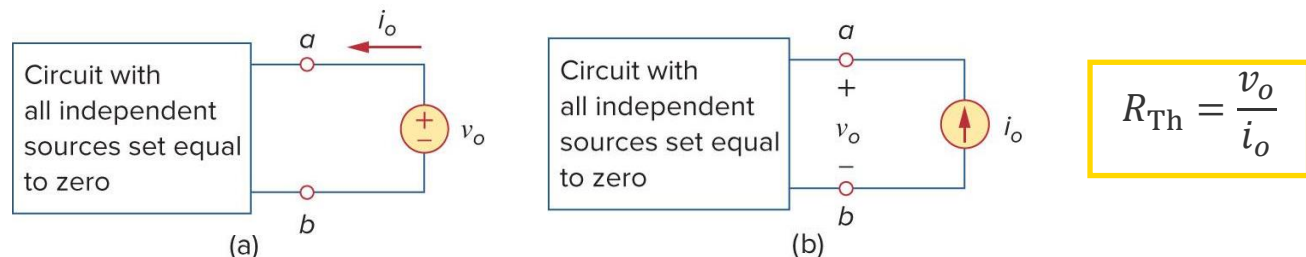
- R_{Th} can be obtained using different methods:
 1. Input **resistance** measured at the terminal pair when **all independent sources are turned off** (not valid if there is any dependent source).



2. Ratio of the open-circuit voltage to the short-circuit current at the terminal pair (not valid if there are only dependent sources).



3. Turn off independent sources and (a) attach a voltage source v_o to the terminals a - b and find the resulting current i_o , or (b) attach a current source i_o and find the resulting voltage v_o . For simplicity: $v_o = 1$ V or $i_o = 1$ A (valid always).



Topic 3 recap

- Norton's theorem.
 - A linear circuit can be **replaced** (modelled) with a **current source** I_N in **parallel** with a **resistor** R_N from a given terminal.
 - I_N is equal to the **short-circuit current** at the terminals ($I_N = i_{sc}$).
 - R_N is the **same** as R_{Th} .
- Thevenin-Norton transformation is exactly the **same** as **source transformation**.
 - $V_{Th} = R_N I_N$ or $I_N = \frac{V_{Th}}{R_{Th}}$
 - $R_{Th} = R_N$
- Maximum power transfer
 - If $R_L = R_{Th}$ the power transferred to the load R_L is maximum.
 - $p_{max} = \frac{V_{Th}^2}{4R_{Th}}$
 - $v_L = \frac{V_{Th}}{2}$